

**VYTAUTAS MAGNUS UNIVERSITY**  
**FACULTY OF INFORMATICS**

**SOCIAL MEDIA ANALYTICS**

**Lab work #2**

**Visualization of Virtual Communities and Explanation of Results**

**Email-Eu-core Network**

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## 1. Aim of the Work

The aim of this laboratory work is to identify and visualize virtual communities in a social network and explain the resulting community structure. The analysis focuses on detecting groups of actors with relatively strong internal connections, identifying important actors and connections between communities, and interpreting the detected groups using information provided with the original dataset.

## 2. Dataset and Network Overview

The Email-Eu-core dataset was obtained from the Stanford Large Network Dataset Collection (SNAP). It represents email communication between members of a European research institution. Each node represents an individual member of the institution, while a directed edge represents an email relationship between two members, with the direction indicating the sender and recipient. The original network contains 1,005 nodes and 25,571 directed email relationships. A separate file provides department labels for all 1,005 nodes, which provides additional organizational context for interpreting the detected communities.

The dataset was selected because email communication represents interaction between actors and can therefore be studied as a social network. The department metadata also makes it possible to compare communities detected from network structure with organizational structure.

Original dataset: <https://snap.stanford.edu/data/email-Eu-core.html>

## 3. Tasks and Practical Work

### *Task 1: Community Detection Method*

The analysis was performed in Python using the NetworkX library. Community detection was carried out using the Louvain method. Community detection identifies groups of nodes with relatively stronger connections within groups and weaker connections between groups.

The original email network is directed because an email has a sender and a recipient. For Louvain community detection, the network was converted to an undirected graph so that the analysis focused on whether actors were connected rather than on the direction of individual email messages. The original directed network contains 25,571 edges, while the undirected network used for community detection contains 16,706 edges. The reduction occurs because reciprocal relationships between the same pair of actors are represented as a single undirected connection.

The network was visualized using a force-directed layout. Community membership was represented using colors, and node size was scaled according to degree so that highly connected actors were more visible. Because the complete network contains 1,005 nodes, an additional community-level visualization was created. In that view, each node represents a detected community, its size represents the number of members, and edge thickness represents the number of connections between communities.

### *Task 2: Community Detection Results*

The Louvain method identified 28 communities. The communities vary considerably in size. The largest detected community contains 251 nodes, followed by communities containing 145, 129, 129, 111, and 93 nodes. Other larger communities contain between 10 and 64 nodes, while 19 detected communities consist of a single node.

The visualization of the nine communities containing at least ten nodes provides a clearer view of the main community structure. These communities form a large interconnected central structure rather than appearing as completely isolated groups. Several communities have substantial connections with one another, while the single-node communities are peripheral or have limited visible interaction in the simplified view.

### *Task 3: Important Actors and Community Structure*

The community-level analysis identified 36 connections between the 28 detected communities. The strongest observed connection is between Community 8 and Community 28, with 815 connections in the undirected network. Other strong connections include Community 2 and Community 8, with 801 connections, and Community 3 and Community 8, with 725 connections. These results show that the detected communities are interconnected rather than completely separated. Community 8 is particularly well connected to several other communities.

Important actors were examined using degree and betweenness centrality. Degree represents the number of direct connections, while betweenness centrality indicates how often an actor lies on shortest paths between other actors. Node 160 had the highest degree, with 347 direct connections, and the highest betweenness centrality, with a value of 0.087415. It belongs to Community 6 and has connections spanning nine detected communities. Other highly connected actors include Nodes 121, 82, 107, and 86.

Community size and connectivity differences are visible in the analysis, but community density was not calculated separately. Therefore, no direct claim is made about which community is the densest.

### *Task 4: Type and Meaning of the Virtual Community*

The Email-Eu-core network is most closely related to an in-group community. The course describes in-group networks as potentially occurring among professional peers in a work environment. The Email-Eu-core network fits this interpretation because the actors are members of the same research institution and the detected groups are formed from their communication patterns. However, the network does not perfectly represent an in-group structure because several detected communities have substantial connections with one another.

The department labels provide additional evidence for interpreting the communities. Community 4 contains 93 nodes, of which 88 (94.62%) belong to Department 14. Community 3 contains 129 nodes, with 93 (72.09%) belonging to Department 4, and Community 1 contains 64 nodes, with 43 (67.19%) belonging to Department 1. In contrast, Community 8, the largest community with 251 nodes, has only 47 members from its largest department group (18.73%). This indicates that Community 8 is more mixed and does not correspond closely to a single department.

The results suggest that some detected communities may reflect organizational departments or groups of closely interacting colleagues, while others may reflect communication patterns that cross department boundaries. The network provides evidence about interaction patterns, but it does not establish why these communities exist or what was discussed in the emails.

#### *Task 5: Connections Between Communities and Limitations*

The cross-community connections indicate that communication occurs between the detected groups. Several actors have connections spanning nine different communities, with Node 160 being particularly notable because of its high degree and betweenness centrality. Such cross-community actors can occupy bridge-like positions in the network and may contribute to interaction between otherwise more separated groups.

The network structure can therefore support an interpretation of potential information flow between communities. However, the network alone cannot establish what information was exchanged, why actors communicated across communities, or whether a connection represents a strong social relationship.

Several limitations should be considered. An email relationship does not necessarily indicate the strength or nature of a social relationship. Louvain communities are produced from network structure and should not automatically be interpreted as formal organizational departments. The full visualization also simplifies a relatively large network, making some individual relationships difficult to observe. Finally, the analysis describes observed network structure but does not establish why particular communities or communication patterns exist.

## **4. Conclusion**

The Email-Eu-core analysis identified 28 communities in a network of 1,005 nodes and 25,571 original directed email relationships. The communities vary substantially in size and are interconnected through 36 community-level connections. Node 160 was identified as a particularly important actor based on its degree, betweenness centrality, and connections across nine communities. Department metadata showed that some communities are strongly associated with particular departments, while others are more mixed. Overall, the analysis demonstrates how community detection, network visualization, centrality measures, and node metadata can be combined to interpret virtual community structure while recognizing the limits of what the network data can establish.

## **Reference**

Stanford Large Network Dataset Collection (SNAP). Email-Eu-core dataset.  
<https://snap.stanford.edu/data/email-Eu-core.html>